MIL-L-12997B
12 December 1958
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION
LINK, CARTRIDGE, METALLIC BELT; 20MM, M12

1. SCOPE

1.1 This specification covers metallic belt links intended to hold 20MM ammunition in the form of flexible belts.

1.2 Classification. - Links shall be of the following types:

Type I - Welded
Type II - Non-welded (alternate)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111B - Tape, Paper, Gummed (Sealing and Securing)
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner
PPP-B-636 - Boxes, Fiber

MILITARY

MIL-G-14461 - General Quality Control Requirements for Ordnance Material
MIL-P-116 - Preservation, Method of
MIL-G-2550 - General Specifications for Ammunition Except Small Arms Ammunition
MIL-C-12968 - Coating, Phosphate, Protective (For Iron and Steel)
MIL-B-13239 - Barrier Material, Waterproof, All Temperature, Flexible
MIL-P-3420 - Packaging Material, Volatile Corrosion Inhibitor Treated
MIL-L-10547 - Liners, Case, Waterproof
MIL-W-12332 - Welding, Resistance, Spot and Projection: For Fabricating Assemblies of Low Carbon Steel

FSC 1305

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STANDARD

FEDERAL

Federal Test Method Std-151

MILITARY

MIL-STD-10 - Surface Roughness, Waviness and Lay
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

ORDNANCE CORPS

D7147028 - Link, Metallic, 20MM, M12 (T61E3)
D7548085 - Link, Metallic, 20MM, M12 (T61E3) Alternate
GL7101289 - Index to List of Inspection Gages and Parts Gaged on Link, Metallic, 20MM, M12 (T61E3)

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. -Metallic Belt Links, Type I (welded), furnished under this specification shall conform to Drawing D7147028 and the drawings and specifications referenced thereon. Metallic Belt Link, Type II (non-welded) furnished under this specification shall conform to Drawing D7548085, and the drawings and specifications referenced thereon.

3.2 The steels used in the manufacture of the links shall be as specified on the drawing.

3.3 Hardness. -Links shall be heat treated to produce a hardness within the hardness range specified by the drawings. Heat treatment processes other than oil quenching and tempering shall not be performed by the contractor without prior approval of the responsible procuring activity.

3.4 Cartridge retention. -Completed links shall meet the retention test requirements as specified on applicable drawing.

3.5 Pull. -The assembled links shall be capable of meeting the pull test requirements as specified on applicable drawing.

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3.6 Visual defects. - Finished links shall be free of burrs, cracks and malformations.

3.7 Final protective finish. - Following heat treatment, the links shall be treated for corrosion protection as specified herein and on the drawing.

3.7.1 Primary coating.

3.7.1.1 Type of coating. - The primary coating shall be a phosphate coating conforming to Type A, or Type B, Class 1 of Specification MIL-C-12968.

3.7.1.2 Continuity of coating. - The primary coating shall be continuous over all surfaces except that the absence of the coating on edges shall be permissible.

3.7.1.3 Procedure. - The primary coating shall be applied to the component parts of the link prior to assembly of the link.

3.7.2 Supplementary coating. - The supplementary coating shall be a thin film preservative compound conforming to Type P-9 of Specification MIL-P-116. The compound shall be applied by dipping, with the excess compound removed by draining or centrifuging.

3.7.3 Salt spray. - The links shall show no signs of corrosion products when tested in accordance to MIL-C-12968.

3.7.4 Welding. - (Applicable to Type I links only) Resistance welding of the stress bridge shall be performed prior to heat treatment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor Quality Assurance System. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall not be restricted to a particular quality assurance system. However, the system adopted must be one which consistently results in material meeting the requirements as evidenced by the evaluation specified in 4.1.2.1. In all testing inspection the use of the equipment specified and the methods of test prescribed in this specification and other documents referenced herein shall be mandatory. Results of all examinations and tests performed under this quality assurance system shall be made available to the Government. The contractor shall notify the Government of and obtain approval for any change to the written procedure that affects the degree of assurance required by this specification and other documents referenced herein.

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4.1.1.1 Submission of data.-At the time the product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance pro visions and requirements as required by the contractual documents.

b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Drawing and specification numbers and dates, together with an identificat ion and date of changes.

e. Number of items in lot.

f. Certificates of analysis including results of tests on materials used in the lot.

g. Date submitted.

4.1.1.2 The certificate shall be signed by a responsible agent of the certifying organization and shall be substantiated by evidence of the agent's authority to bind his principal.

4.1.2 Government Verification.

4.1.2.1 At the beginning of the contract and until such time as the contractor has operated under an established and approved quality assurance system which has resulted in an acceptable product, Government verification will consist of (l) performance of all inspections and tests listed in the detail specification and other documents referenced therein, which are not reserved for sole performance by the contractor, (2) monitoring of those inspections reserved for sole performance by the contractor, (3) monitoring of the contractor's quality assurance operations. At no time will complete Government verification be reduced less than that permitted by MIL-STD-105A and ORD-M608-11 except under those conditions prescribed in 4.1.2.2.

4.1.2.2 Under the following conditions further reduced Government inspection may be initiated on a complete or individual operation basis.

a. On a voluntary basis the contractor includes in his quality assurance provisions written quality control procedures evaluated and approved by the Government.

b. Verification inspection by the Government in accordance with 4.1.2.1 indicates that the quality of product continues to be equal to or better than that required by this specification and the referenced documents.

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4.1.2.3 At such time as the contractor meets the conditions of 4.1.2.2 above, Government verification will consist of, as a minimum, inspection for critical defects, monitoring of contractor's quality assurance system for strict compliance to the written plan and review of contractor records on all inspection performed. If the contractor's quality deteriorates or a departure from the written plan is noted, the Government inspector will immediately notify the contractor to take corrective action and until the deficiencies have been corrected, Government verification in accordance with 4.1.2.1 will be performed on those characteristics involved to determine acceptability of the product. If corrective action is not taken by the contractor, as requested by the Government, acceptance of the product may be suspended. Upon correction of the deficiencies, the Government will continue inspection as prescribed in 4.1.2.1 until contractor has re-qualified for reduced Government verification as prescribed in paragraph 4.1.2.2.

4.1.2.4 Critical Defect. -Government inspection for critical defects will be conducted at all times in accordance with MIL-G-2550.

4.2 Preproduction sample. -Prior to the start of regular production, the contractor shall submit a preproduction sample of 300 links manufactured by the same method and equipment which will be used in performing the work under contract. The contracting officer will furnish instructions for shipment of the sample to a Government Laboratory for examination and test. The samples will be inspected for all the requirements specified on the applicable drawings and in this specification. The preproduction sample shall consist of:

Thirty-five (35) samples selected for Hardness test.

Seventy-five (75) sample links shall be selected for the cartridge retention test.

Seventy-five (75) sample links shall be selected for the Pull test.

One-hundred (100) links shall be selected for visual and gage. These links shall be gaged 100% on all dimension:

Ten (10) links will be available for coating tests.

Five (5) links will be available for weld test. (Type I link)

4.2.1 Results of the examination and test will be furnished to the contractor. Approval or disapproval of the preproduction sample will be given the responsible testing activity. Instructions for the submission of new samples in case of disapproval of the original will be issued by the testing activity. Regular production shall not proceed until the preproduction sample has been approved.

4.3 Acceptance inspection.

4.3.1 Lot formation. -Unless otherwise specified a lot shall consist of links produced by one manufacturer in one unchanged process, according to the same drawing, same drawing revision, same specification, and same specifica-

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tion revision, and in addition a lot shall contain:

a. Not less than 22,000 links
b. Links from not more than 1 heat of steel

4.3.2 Examination inspection. -Examination shall be performed as specified in Specification MIL-G-2550 and as indicated below. Samples and acceptance numbers for the inspection specified in the applicable drawing will be in accordance with Standard MIL-STD-105 or Publication ORD-M608-11 as appropriate.

4.3.2.1 The defect shall be used as a basis for judgement using the following AQL's:

Major defects	0.25% for each defect
Minor defects	0.40% for each defect

4.3.2.2 The contractor shall examine each unit for all critical defects listed in 4.3.4.

4.3.3 Inspection testing. -Samples for the following tests will be selected by the inspector.

4.3.3.1 Hardness. -Thirty-five (35) samples shall be selected from each heat treat batch or shifts production and tested for compliance with the drawing requirements for hardness in accordance with the test specified on the drawing. If any links are found which do not meet the drawing requirements, the production represented by the sample shall be rejected.

4.3.3.2 Cartridge retention test. -Seventy-five (75) sample links shall be selected from each shifts' production and assembled for testing as specified in 4.4.2. If three or more defects are found, a retest may be conducted using 150 samples. If the number of defects from both tests combined is three or more, the shifts' production shall be rejected.

4.3.3.3 Pull test. -Seventy-five (75) sample links shall be selected from each shifts' production and tested as specified in 4.4.3. If three or more defects are found, the shifts' production shall be rejected. If one or two defects are found, a retest may be conducted using 150 samples. If the number of defects from both tests combined is three or more, the shifts' production shall be rejected.

4.3.3.4 Primary coating.

4.3.3.4.1 Continuity of Coating. -Continuity of coating shall be tested by the salt spray test in accordance to MIL-C-12968. If one link exhibits rust discoloration, the links represented by the sample shall be rejected.

4.3.3.5 Weld test. -Welded links shall be tested in accordance with MIL-W-12332. The quantity of links to be submitted to the spot weld test shall be as specified:

Sampling for welding (Peel) test of links
Hourly sample size per machine - 6
Retest sample size (same machine) - 20

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Acceptance is based on the following:

1st Sample	Batch Status
Failure of no links	Acceptance
Failure of one link	Retest from same batch
2nd (Retest Sample)	
Failure of no links	Acceptable
Failure of one link	Unacceptable

4.3.4 Classification of defects.

4.3.4.1 Links.

Critical defects	Method
a. Malformed links	Visual
b. Cracks in critical areas	Visual
Major defects	
1. Burrs	Visual
2. Width; flared end stress arm (w/ring)	Gage
3. Width; end, stress arm (w/o ring) Type I link only	Gage
4. Width; end, stress arm (w/o ring) Type II links only	Gage
5. Grip on cartridge (min opening)	Gage
6. Max clearance between bridge or body arms	Gage
7. Min distance from bridge arms or body arms to link vertical C. L. (w/steel plug in link)	Gage
8. Min distance from tip of flared end of body stress arm (w/ring) to longitudinal center line of link w/plug in link	Gage
9. Height of bridge trip from bridge base, Section BB of assembly drawing	Gage
10. Width; tip of stress bridge (for non-welded links only)	Gage
11. Guide tabs offset height; inner surface, max. and outer surface, min. (gage w/steel plug in link before retention test	Gage
Minor defects	
41. Distance between extreme inside width of body stress arms	Gage
42. Location: front end of front guide tab (w/steel plug in link)	Gage
43. Overall length; guide tabs	Gage
44. Maximum overall length; longitudinal offset of guide tabs (w/plug in link)	Gage
45. Minimum distance; front surface of ramming guide embossment to rear surface of rear guide tab longitudinal offset/figured dimensions	Gage

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4.3.4.1 Links (Contd)

Minor defects (contd)	
46. Overall length of links	Gage
47. Inside diameter of connector ring	Gage
48. Clearance; feeder channel guide	Gage
49. Improper or missing identification	Visual

4.3.4.2 Packaging

Major defects	Method
1. Marking missing or misleading	
2. Container punctured, crushed or torn	
3. Container interior contaminated with dirt, lint, or other foreign matter	
4. Container improperly sealed or closed	
5. Container packing, parts, i.e., partitions, pads, separators missing, torn or crushed	
Minor defects	
41. Markings incorrect or incomplete	
42. Improper oil application and drainage	

4.3.5 Examination equipment

4.3.5.1 Government verification inspection. -Inspection will be accomplished with the use of gages and equipment that has been certified, by the contracting officer, to the design specified in the Index of Inspection Equipment Lists GL-7101289.

4.3.5.2 Contractor inspection. -Unless otherwise specified herein contractor inspection will be accomplished with the use of gages and equipment designed to perform, to the desired accuracy, the inspections and controls delineated in the contractor's written description specified in 4.1.1. In inspecting the contractor's inspection equipment the Government inspector will determine that the contractor has available and utilized correctly, gaging, measuring and test equipment of required accuracy and that the instruments are of proper type and range to make measurements of the required accuracy.

The contractor shall have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Ordnance personnel.

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4.4. Test procedure. -The following tests will be performed by the contractor.

4.4.1 Hardness. -Each link of the sample shall be tested for compliance with drawing requirements for hardness. The test method shall be performed with Rockwell Hardness Testers or equivalent. All hardness will be conducted prior to phosphating. The average of two reading per link tested shall be considered as the hardness of the part.

4.4.1.1 Type I links. -Welded links shall be tested for hardness on top of the stress bridge outside the welded area. Samples shall be taken from link bridges removed during the welding test.

4.4.1.2 Type II links. -Non-welded links shall be tested for hardness after heat treatment by conducting seperate tests on the bridge and body.

4.4.2 Cartridge retention test. -Each link of the sample shall be tested for compliance with 3.4 when tested with the applicable Ordnance test fixture as listed in GL7101289. The links shall be assembled and the method of use as described on the fixture drawing shall be followed. The mandrel shall be made of steel of the minimum projectile outside dimensions. Partial extraction shall be classed as failure. (See note 6.2.5)

4.4.3 Pull test. -Each link of the sample shall be tested for compliance with 3.5 when tested with the applicable Ordnance Test Fixture and equipment as listed on GL7101289. The method of use as described on the fixture drawing shall be followed. The links shall be inserted in the fixture and subjected to the required load without fracture or without exceeding the distance x-x of Drawing 7147028 or 7548085 as applicable.

4.4.4 Weld test. -Each link of the sample shall be tested for compliance with the drawing. The links shall be tested in accordance with MIL-W-12332.

4.5 Acceptance

4.5.1 At the beginning of the contract and as long as the Government inspector continues to perform the verification specified in 4.1.2.1, acceptance will be based on the results of the Government verification specified in 4.1.2.1 and 4.1.2.4.

4.5.2 After reduced Government verification specified in 4.1.2.2 has been initiated, acceptance will be based on the results of the contractor's inspection and quality control and the inspection reserved solely for performance by the Government and the results of the monitoring by the Government of the contractor's quality assurance program.

5. PREPARATION FOR DELIVERY

5.1 General. -Unless otherwise specified in the contract the level of protection for shipment shall be in accordance with Level C.

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5.2 Pilot Pack. -On each contract a pilot pack consisting of a complete and packed shipping container representing initial production shall be taken by the Government inspector and shall be forwarded f.o.b. contractor's plant to a testing agency designated by the procuring activity. (See 6.1) The pilot pack shall be representative of material which has been inspected and approved by the Government inspector. The pilot pack shall contain the production samples of links.

5.3 Preservation and packaging.

5.3.1 Level A

5.3.1.1 Cleaning. -The links shall be cleaned by process C-3 of Specification MIL-P-116. In those cases where the links go directly from the supplementary wax coating of the final protective finish process to the packaging operation without intermediate handling, such as inspection, the links may be packaged and packed without cleaning, drying and reoiling or rewaxing, provided the links are tested and found free of contamination.

5.3.1.2 Drying. -Links that are cleaned by process C-3 of Specification MIL-P-116, shall be dried in accordance with any applicable drying procedure of Specification MIL-P-116.

5.3.1.2.1 Rewaxing. -The supplementary oil or wax coating of the final protective finish shall be reapplied to the cleaned links. Excess supplementary coating shall be thoroughly removed by draining or centirfuging prior to packaging.

5.3.1.3 Unit packaging. -Five hundred links shall be unit packaged in a Class 1, Specification PPP-B-636 fiberboard box. The links shall be coupled together to form belts of 10 links each. The belted links shall be packaged in horizontal rows, five rows of ten links each per layer, ten layer deep. Each layer of links shall be separated by VCI treated pads conforming to Type I of Specification MIL-P-3420. The pads shall be VCI treated on both sides. The box shall be of the minimum dimensions necessary to contain the links and all joints and seams of the box shall be sealed with 3-inch tape conforming to Specification UU-T-111B.

5.3.2 Level C.

5.3.2.1 Cleaning, drying and preservation. -Cleaning, drying and preservation shall be in accordance with 5.3.1.1, 5.3.1.2, and 5.3.1.2.1.

5.3.2.2 Unit packaging. -Links shall be packaged in accordance with 5.3.1.3 except that fiberboard box shall conform to the requirements of Class I of Specification PPP-B-636.

5.4 Packing.

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5.4.1 Level A. -Two unit packages shall be packed in Class 2, Style 2, nailed wood box conforming to specification PPP-B-621. A case liner conforming to Type I, Grade A or B, specification MIL-L-10547 constructed of material conforming to Type CW-1 or CW-2 of Specification MIL-B-13239 shall be used within the shipping container. Strapping of the container shall be in accordance with the appendix of Specification PPP-B-621.

5.4.2 Level C. -The unit package shall serve as the shipping container. The carton shall be closed and sealed in accordance with the appendix of Specification PPP-B-636 to insure acceptance by common carrier and safe delivery at destination.

5.5 Marking. -Unit packages and shipping containers shall be marked with Standard MIL-STD-129.

5.6 Data Cards. -Data card information shall be as specified in MIL-G-2550.

6. NOTES

6.1 Ordering data. -Procurement documents should specify the following:

a. Title, number, and date of this purchase description.
b. Title, number and date of applicable drawings.
c. FSN code number.
d. Level of protection required for shipment if other than specified (see 5.1)

6.2 Definition of terms.

6.2.1 Malformed links. -Links whose shape and/or dimensions have been mutilated or altered by processing to the extent that the links cannot be gaged.

6.2.2 Cracks. -See figures 1, 2a, and 2b.

6.2.2.1 Cracks in critical areas. -Any visible fracture of the link extending through the stock thickness in depth or a distance in length equal to or greater than the material stock thickness.

6.2.2.2 Cracks in non-critical areas. -Any visible fracture of the link extending through the stock thickness in depth but equal to or greater than the stock thickness in length.

6.2.3 Burr. -Any rough edge on any part of the link which, in the opinion of the Government inspector, would produce a definite removal of metal from a cartridge case or which could injure personnel in handling.

6.2.4 Critical area. -The area of the stress arms on the link and the connecting ring shall be designated as critical areas.

6.2.5 Partial Extraction. -When only one end of the mandrel or round becomes free.

6.3 The following documents are superseded by this specification:

RIAPD - 579B
14 September 1955

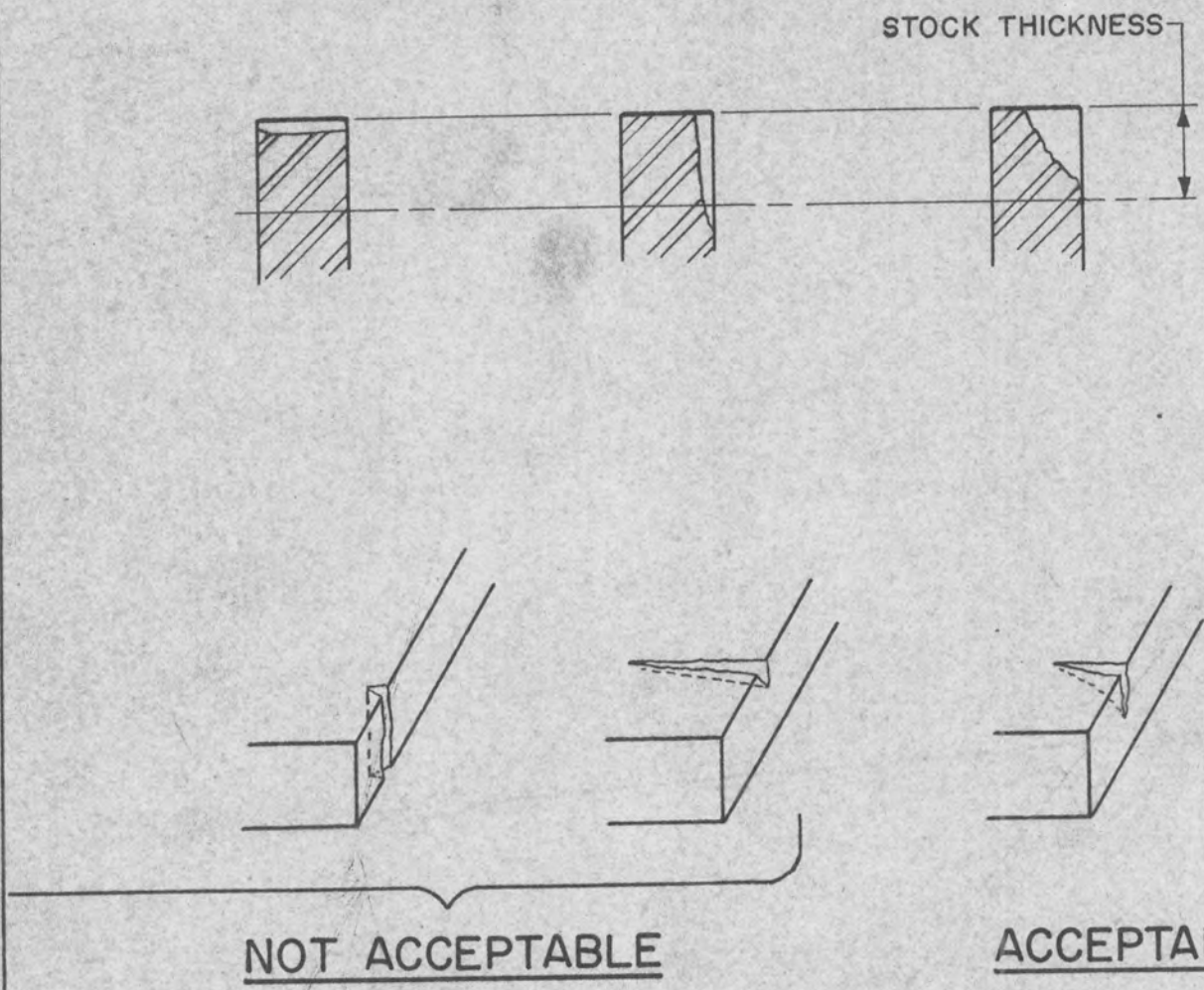
MIL-P-14241 (in Part)
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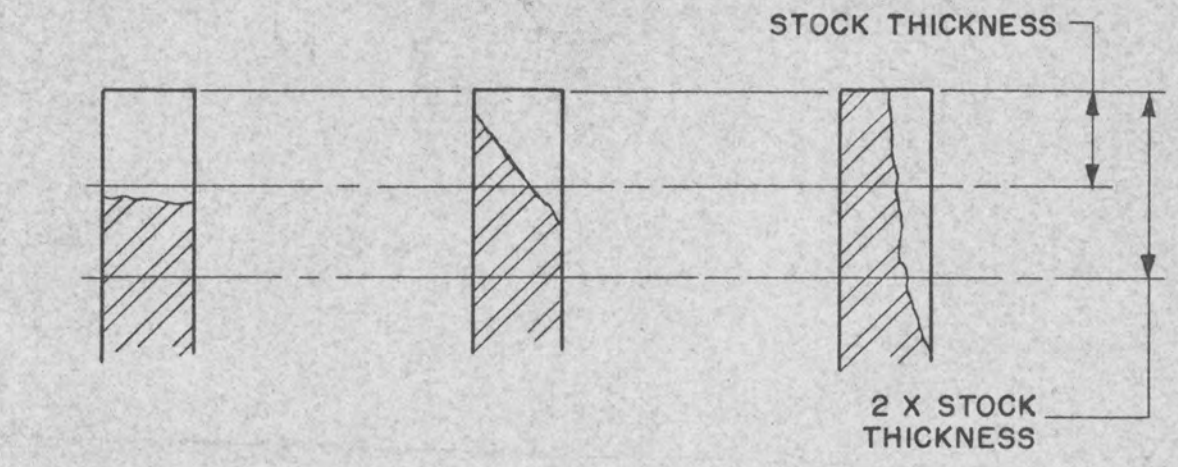
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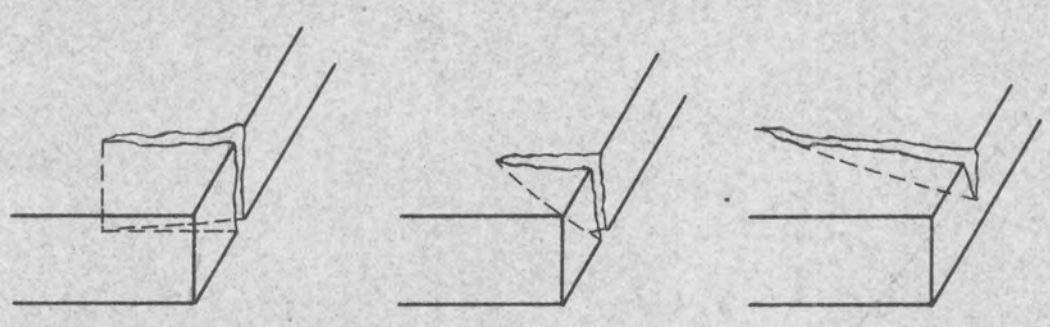


CRACKS IN CRITICAL AREAS

FIGURE 1

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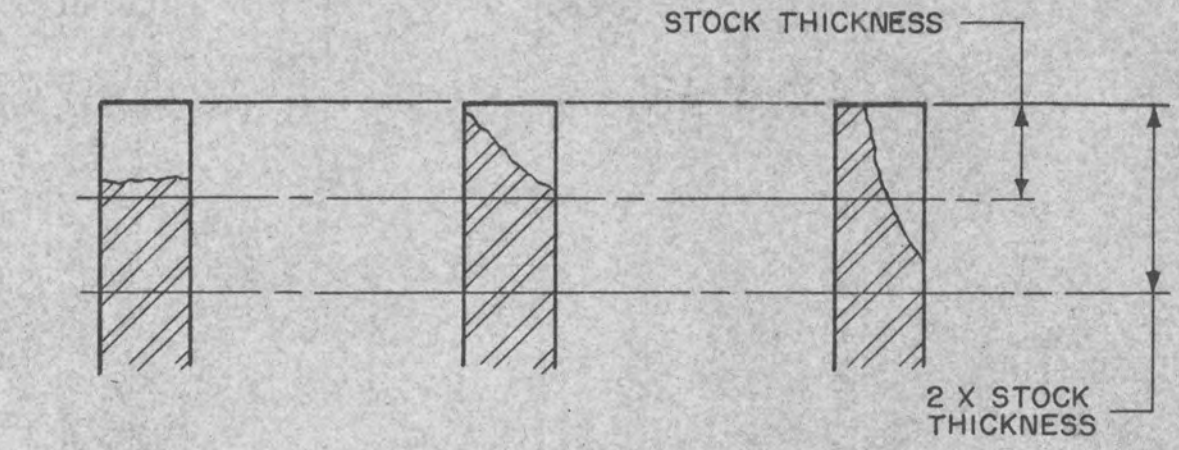


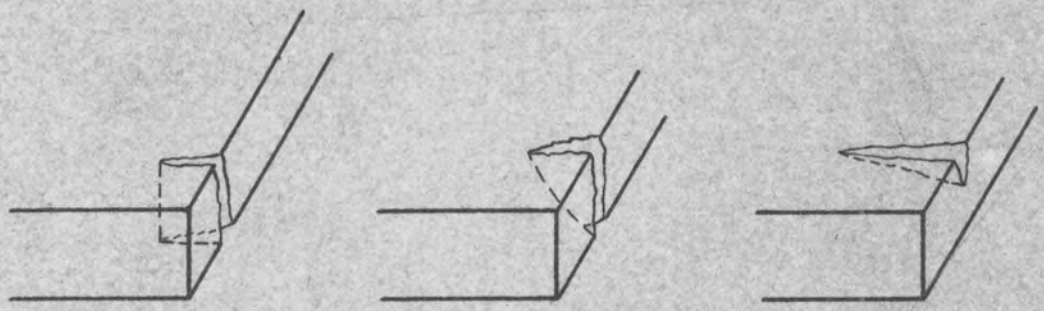
ALL NOT ACCEPTABLE

CRACKS IN NONCRITICAL AREAS

Figure 2 (a)

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CRACKS IN NONCRITICAL AREAS

FIGURE 2 (b)

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